

A Hash-Based Compact Testing Scheme for Peer-to-Peer Chiplet Verification

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Abstract—Chiplets have various advantages in current chip design but also raise piracy and security problems. One key concern is supply chain vulnerability. Because chiplets are frequently acquired from several sources, it becomes challenging to ensure the authenticity of each component. Malicious actors could introduce hardware Trojans into the chiplets, which might be undetected, especially in the absence of standardized security measures across vendors. To address this challenge one must first answer the following question: how can a chiplet instance be tested by a peer chiplet on the fly with limited bandwidth available for testing data? To address this question, we provide a Merkle hash tree testing scheme that utilizes the input-output properties of the circuit and the compression opportunities provided by the hashing algorithm. We performed experiments on complex chiplet design to verify the proposed technique and report the scheme's efficiency using the standard sideband present in the current UCIe protocol[1].

Index Terms—Brute force search, Chiplets, Peer testing

I. INTRODUCTION

The shift from traditional monolithic chips to smaller chiplets combined on a single wafer transforms the way modern semiconductors are built. These chiplets bring clear advantages—better performance, flexibility, and cost savings. However, with this new approach also comes new security concerns. As we make this transition, it is crucial to understand the vulnerabilities that come with it. Our research focuses on exploring these security challenges and finding ways to protect chiplet-based systems from potential threats in this rapidly evolving landscape. This new approach reduces costs, optimizes performance, and minimizes waste. However, this transition also introduces new security challenges, particularly ensuring secure communication between chiplets. During chiplet communication, verifying that the correct chiplet is involved in the exchange is critical to prevent vulnerabilities and unauthorized data access. Several methods are used for data verification, including Merkle hash trees, built-in self-testing (BIST), built-in peer testing (BIPT), SHA-256, the avalanche effect, and automatic test pattern generation.

II. BACKGROUND

A. Merkle Hash Trees

Merkle hash trees[2] play a crucial role in ensuring data integrity and secure communication between chiplets. It is a cryptographic data structure used for data verification. With the help of Merkle trees, small changes are detected and used to check inconsistencies.

B. Built-in Peer Testing (BIPT)

Built-in peer testing (BIPT) extends the scope of testing by enabling peer-to-peer verification, ensuring that interconnected chiplets within a system function correctly and securely.

III. METHODOLOGY

The SHA-256 cryptographic algorithm provides robust encryption by encoding any data size into a fixed 256-bit hash. A key feature in cryptography, known as the avalanche effect, causes substantial output changes even from minor input modifications. This property is crucial for detecting counterfeit chiplets and enhancing security by highlighting tampering risks. In this research, we implement a Merkle hash tree using encoded test vectors (input and output data) for chiplet verification. Input data passes through the D2DSideband module, and the corresponding output helps verify the integrity of a 64-bit multiplier using random test vectors and hash values. Although manufacturers test chips before release, peer testing within chiplets adds a layer of security by performing additional random tests to verify hash values. Chiplet X can only trust Chiplet Y if it returns the correct hash, establishing secure communication.

IV. EXPERIMENTAL SETUP & RESULTS

As part of this project, we successfully implemented peer testing by taking a 64-bit input multiplier and a D2DAdaptersideband module and providing test vectors to the D2DSideband module to test the multiplier. We used six test vectors to generate a final root hash. We are trying to scale the number of test vectors (input and output combinations) to check the power efficiency. We are currently working on that. We have yet to produce results. So, we will submit the results in the final paper submission.

V. DISCUSSIONS

The avalanche effect property of hash functions makes the encoding mechanism more robust. This hash-based compact testing scheme for peer-to-peer chiplet verification makes chiplets communications more authentic. If there is any tampering or trojan hidden inside the chiplet, it can be easily detected. So, this approach should be implemented when testing chiplets for secure communication, which reduces testing costs and builds trust between the chiplets.

REFERENCES

- [1] "UCIe Digital repository, "ucb-ucie"," [Online]. Available: <https://github.com/ucb-ucie/uciedigital>.
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